

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                 | Marks Available |          |          |           |          |          |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                 | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 5        | (a) | (i)  |  | 37 <u>years</u><br>Don't accept 37 light years<br>Accept 1 169 200 000 <u>s</u> or 13 505 <u>days</u>                                                                                                                                                                                                           | 1               |          |          | 1         |          |          |
|          |     | (ii) |  | Substitution: $3 \times 10^8 = \frac{d}{31600\ 000 \times 37}$ <b>ecf</b> (1)<br>Rearrangement: $d = 3 \times 10^8 \times 31\ 600\ 000 \times 37$ (1)<br>$= 3.5 \times 10^{17}$ [m] (1)<br>Award 2 marks for an answer of $9.48 \times 10^{15}$ [m]<br>Award 2 marks for an answer of $1.11 \times 10^{10}$ [m] | 1               | 1<br>1   |          | 3         | 3        |          |
|          | (b) | (i)  |  | <b>Any 2 × (1) from:</b><br>• Arcturus is <u>bigger</u> <b>or</b> has a <u>bigger radius</u><br>• Arcturus is <u>brighter</u> <b>or</b> <u>more luminous</u><br>• Arcturus is <u>cooler</u> <b>or</b> has a <u>lower temperature</u><br>Accept the converse answers.<br>Don't accept reference to 'it'.         |                 | 2        |          | 2         |          |          |
|          |     | (ii) |  | Hydrogen reduces / hydrogen runs out / hydrogen fusion stops (1)<br>Helium fusion begins (1)<br>Increasing gas <b>or</b> radiation pressure (1)<br>As outwards force is larger [than the inwards force] (1)<br>Don't accept Sun has burnt all of its hydrogen                                                   | 4               |          |          | 4         |          |          |
|          |     |      |  | <b>Question 5 total</b>                                                                                                                                                                                                                                                                                         | <b>6</b>        | <b>4</b> | <b>0</b> | <b>10</b> | <b>3</b> | <b>0</b> |